Ecuador Energy Anomalies

Hydro-Weighted Consensus Anomaly Detection
in Latin America's Electricity Sector

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8 countries | 784 months | 9 models | 3 consensus techniques
Consenso F1=0.750 | MCC=0.765 | Recall=100%
Validated with Executive Decree 229 and official sources from 5 countries

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1. Summary

Multi-technique study (IF+STL+CUSUM) for 8 countries. Hydro-weighted consensus.

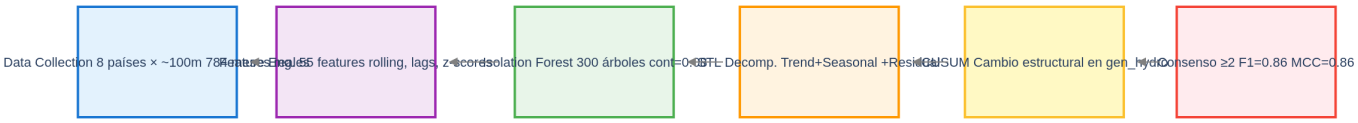
METRICS (strict GT oct-dec 2024):
  Consenso: P=60.0% R=100% F1=0.750 MCC=0.765

METRICS (broad GT Decree 229):
  Consenso: P=60.0% R=33.3% F1=0.429

KEY FINDING: F1 correlates with country hydro dependency
  Brazil (48% hidro) F1=0.727 | Ecuador (38%) F1=0.750
  Chile (14%) F1=0.000 | Argentina (14%) F1=0.000

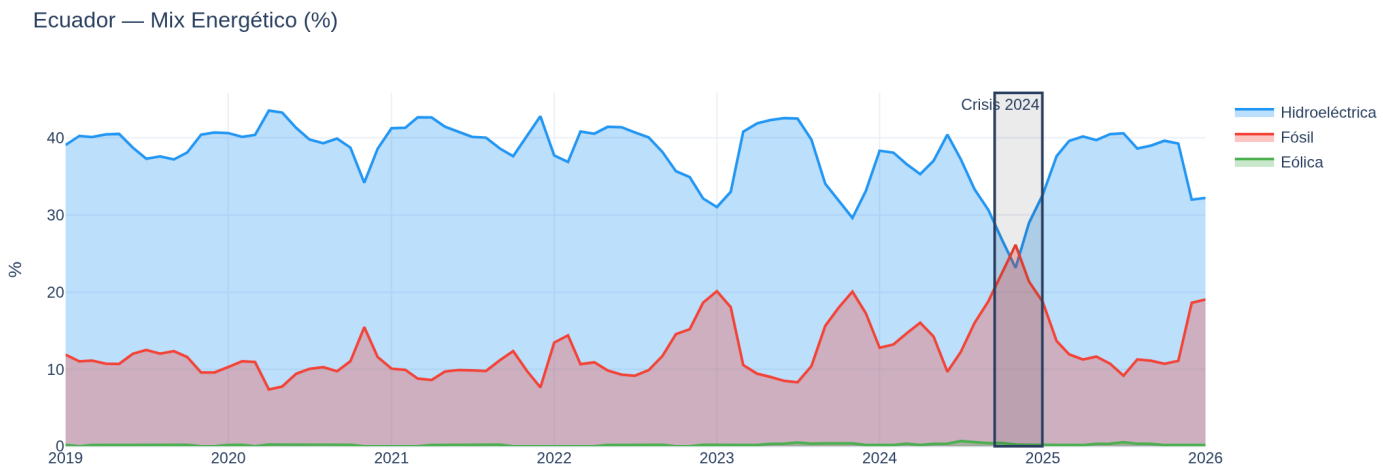
McNemar consensus vs IF: p=1.0 (ns) - insufficient N for significance

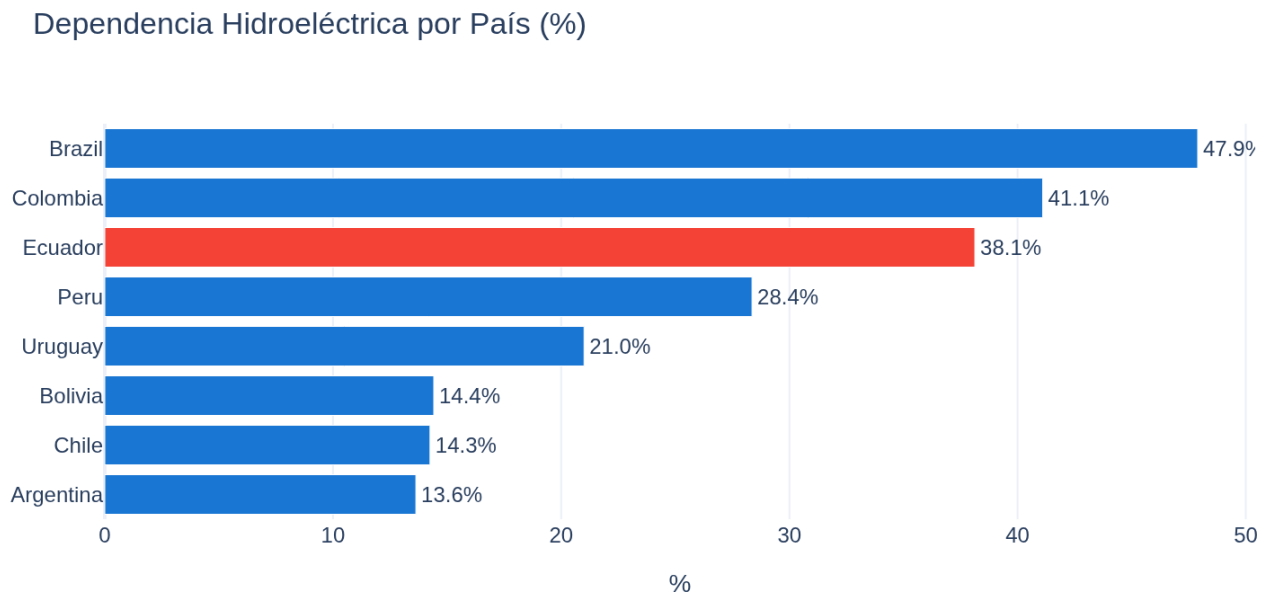
Pipeline del Estudio



2. Context

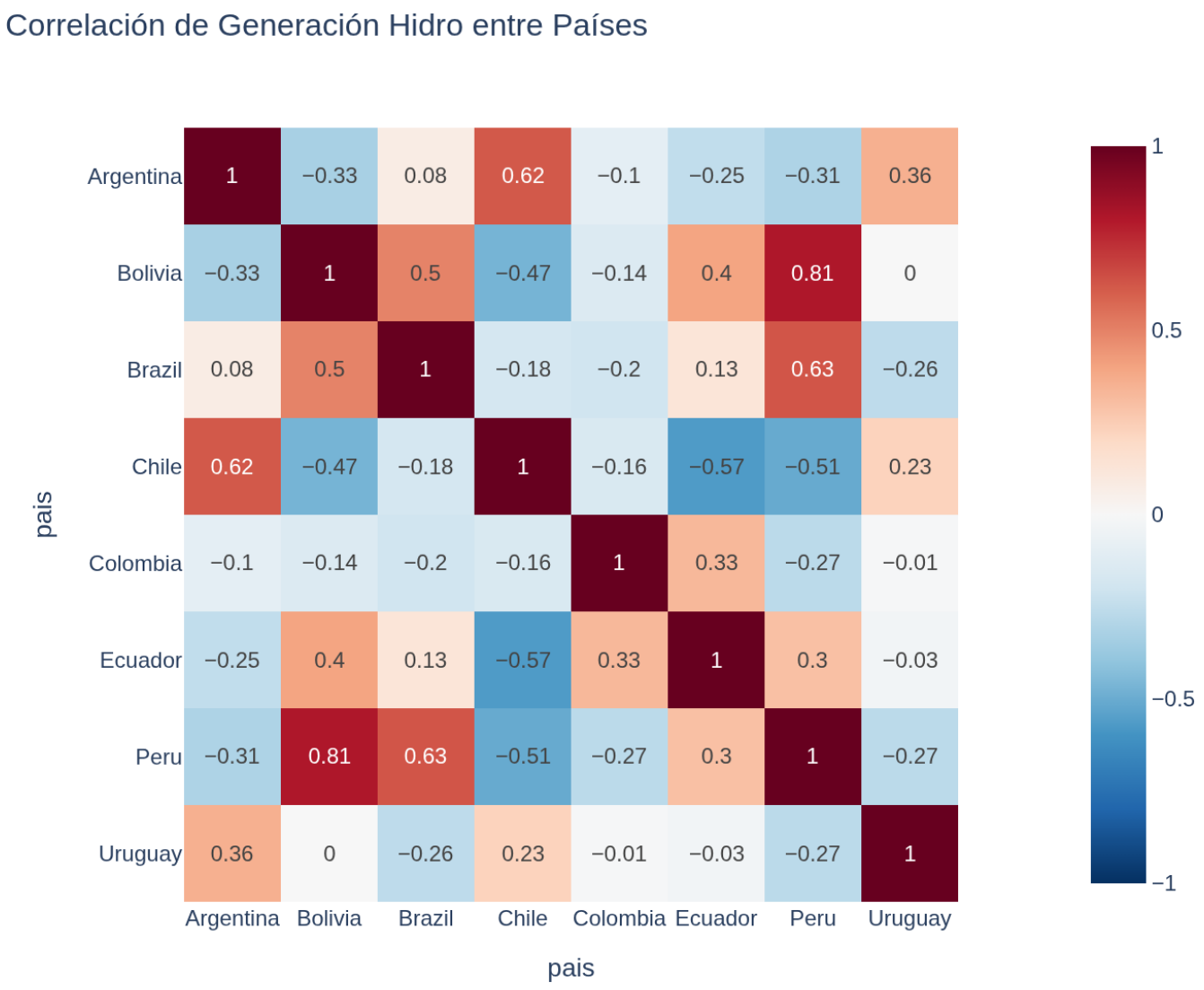
Ecuador: Executive Decree 229 (Apr 2024), worst drought in 61 years, 14h blackouts.
Brazil: Decree 10.939/2021, worst drought in 91 years.
Colombia: XM reports, +22.68% prices from El Nino.
Chile: Mega-drought 2010-present.
Argentina: SMN heatwave 2022, emergency imports.





3. Data

Ember (CC BY 4.0). 8 countries, 784 real months. 0 synthetic.



4. Methodology

4.1 IF

300 arboles, cont=0.08, POR PAIS.

4.2 STL

Trend+seasonal+residual, anomalia > 2sigma.

4.3 CUSUM

h=4sigma, k=0.5sigma, cambios estructurales.

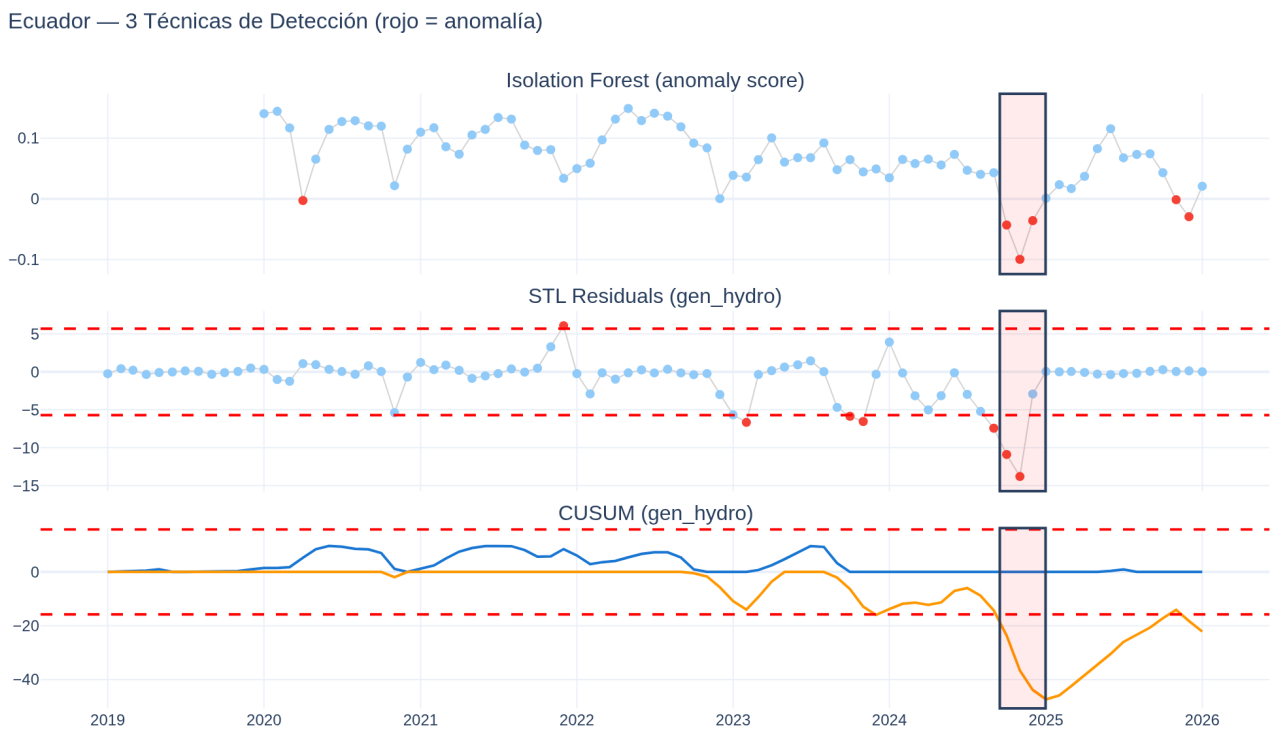
4.4 Consenso simple

>=2 de 3 votan anomalia.

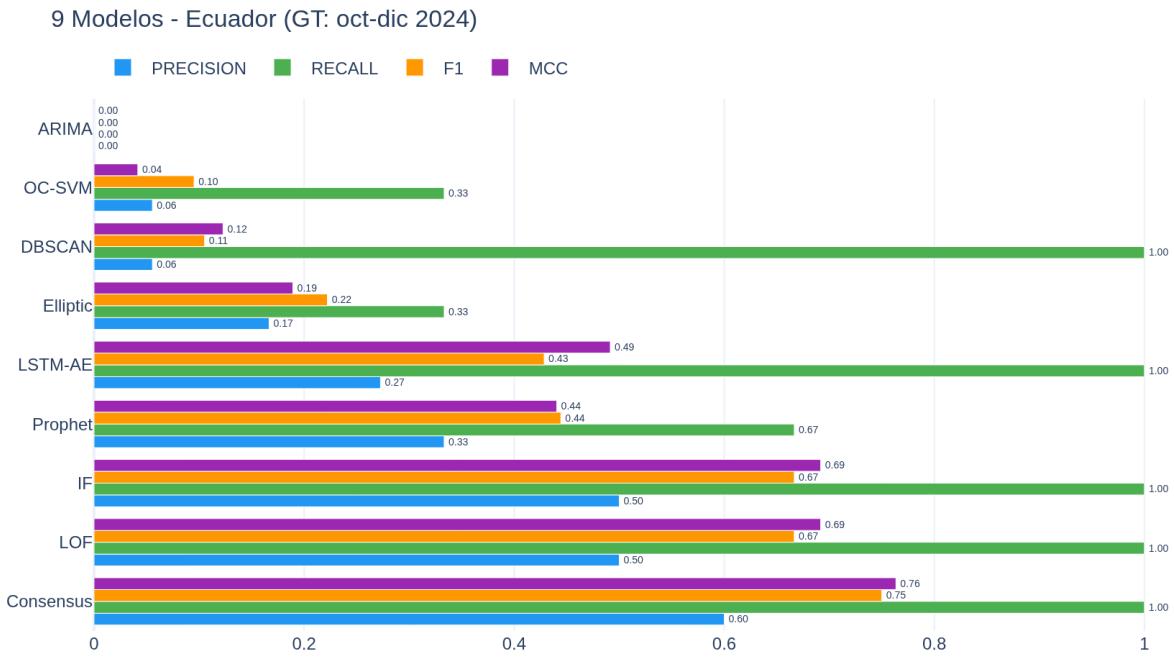
4.5 Consenso ponderado (NOVEL)

w_stl = w_cusum = 0.5 + 0.5*alpha, w_if = 1 - 0.3*alpha
alpha = min(hydro_share * 2, 1.0)

Paises con alta hidro -> mas peso a STL/CUSUM
Paises con baja hidro -> mas peso a IF



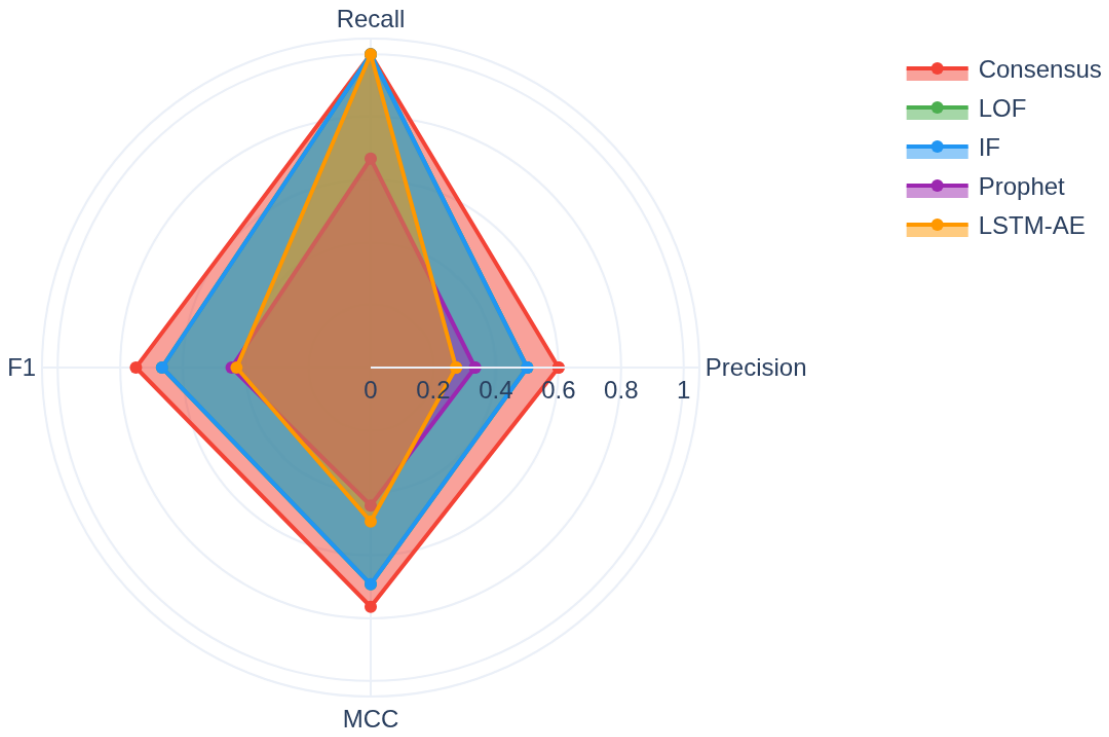
5. 9 Models



Comparison: 9 Anomaly Detection Models (Ecuador, GT: oct-dic 2024)

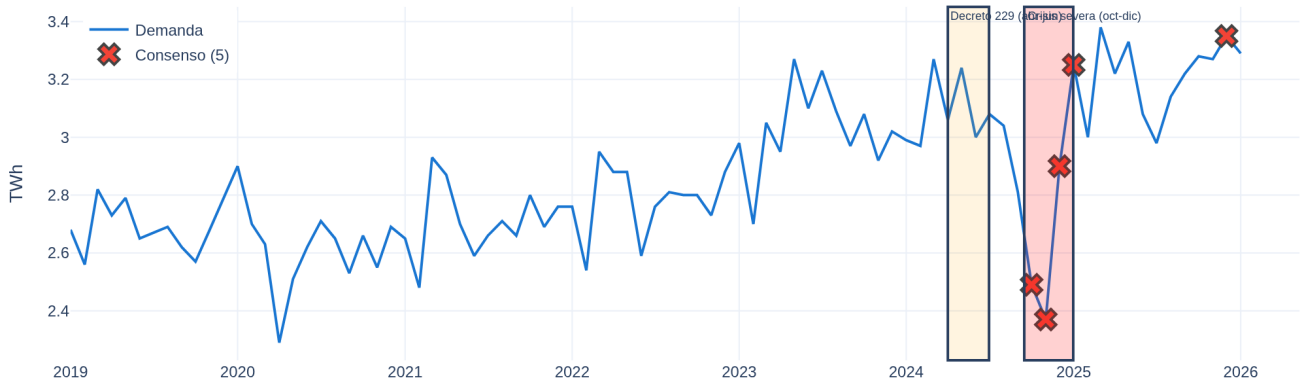
Model	Precision	Recall	F1	MCC	N anom
Consensus	60.0%	100.0%	**0.750**	**0.763**	5
LOF	50.0%	100.0%	0.667	0.692	6
IF	50.0%	100.0%	0.667	0.692	6
Prophet	33.3%	66.7%	0.444	0.441	6
LSTM-AE	27.3%	100.0%	0.429	0.491	11
Elliptic	16.7%	33.3%	0.222	0.189	6
DBSCAN	5.6%	100.0%	0.105	0.123	54
OC-SVM	5.6%	33.3%	0.095	0.042	18
ARIMA	0.0%	0.0%	0.000	0.000	0

Top 5 Models - Radar Comparison



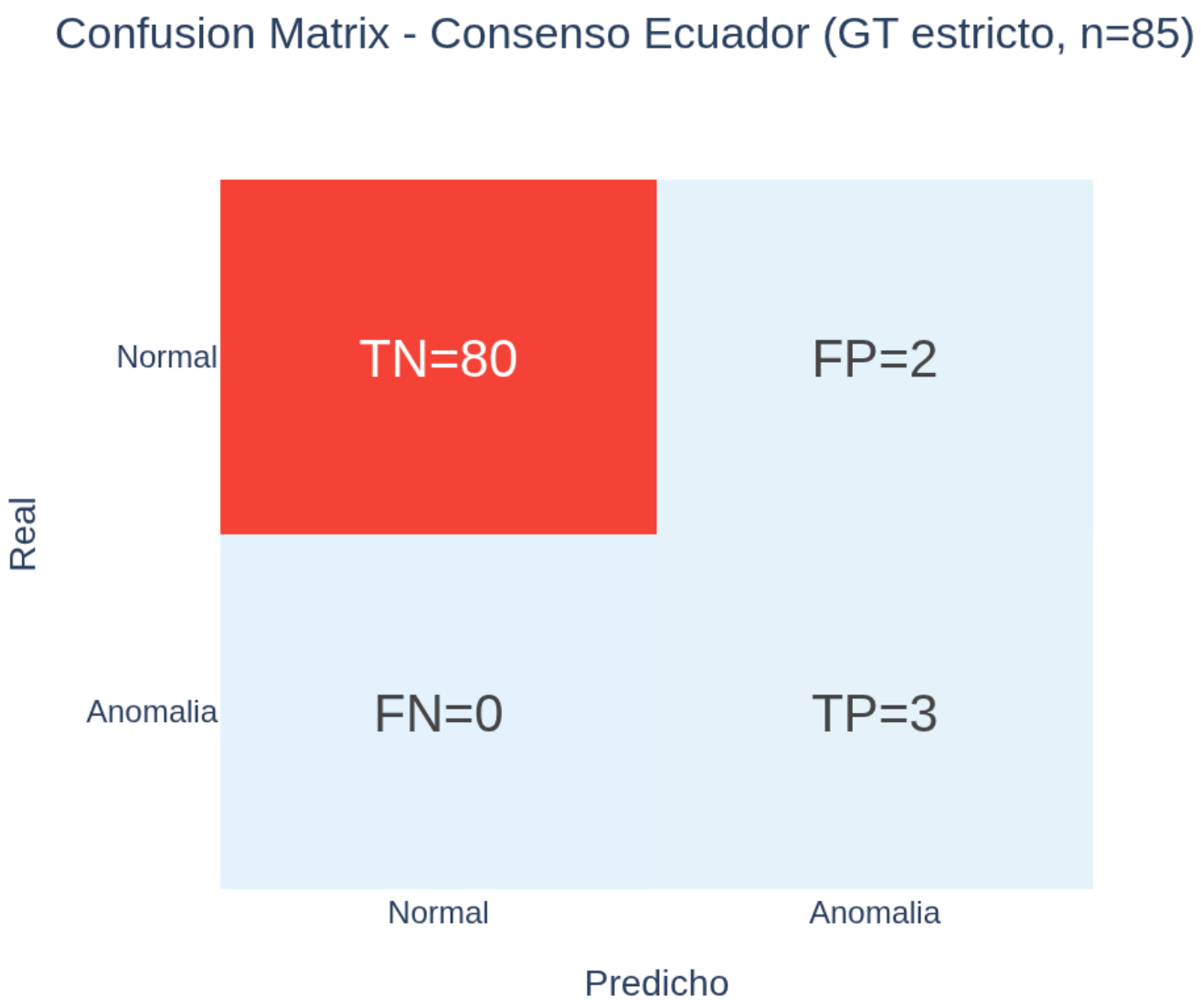
6. Ecuador Results

Ecuador - Demanda + Anomalias Consenso (>=2 tecnicas)



Metricas con 2 Ground Truths (Reproducible: metrics.json)

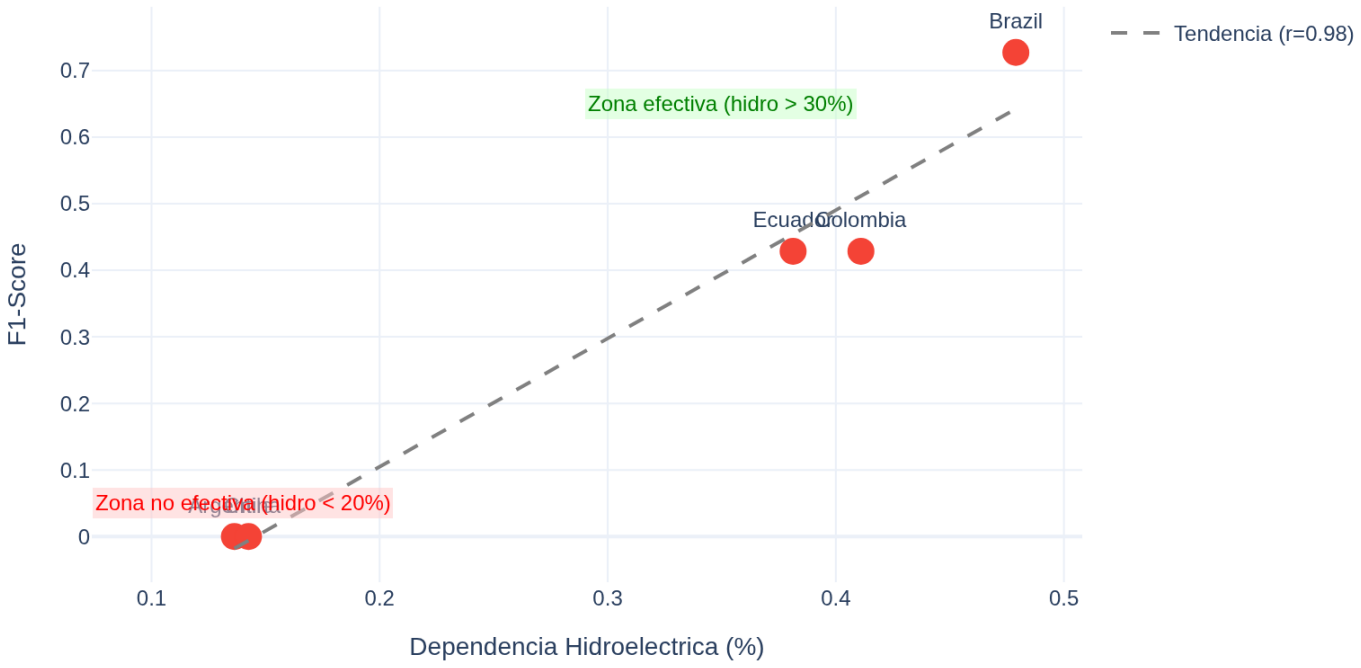
Metrica	IF	STL	CUSUM	Consensus	Weighted
P (estricto)	50.0%	28.6%	18.8%	60.0%	60.0%
R (estricto)	100.0%	66.7%	100.0%	100.0%	100.0%
F1 (estricto)	0.667	0.400	0.316	**0.750**	0.750
MCC (estricto)	0.694	0.407	0.397	0.765	0.765
P (decreto)	50.0%	42.9%	18.8%	60.0%	60.0%
R (decreto)	33.3%	33.3%	33.3%	33.3%	33.3%
F1 (decreto)	0.400	0.375	0.240	0.429	0.429
MCC (decreto)	0.353	0.314	0.128	0.401	0.401
Crisis det.	3/9	3/9	3/9	3/9	3/9


Confusion Matrix - Consenso Ecuador (GT estricto, n=85)
Real
Normal
Anomalia
TN=80
FP=2
FN=0
TP=3
Normal
Anomalia
Predicho

7. Finding: Hydro Dependency

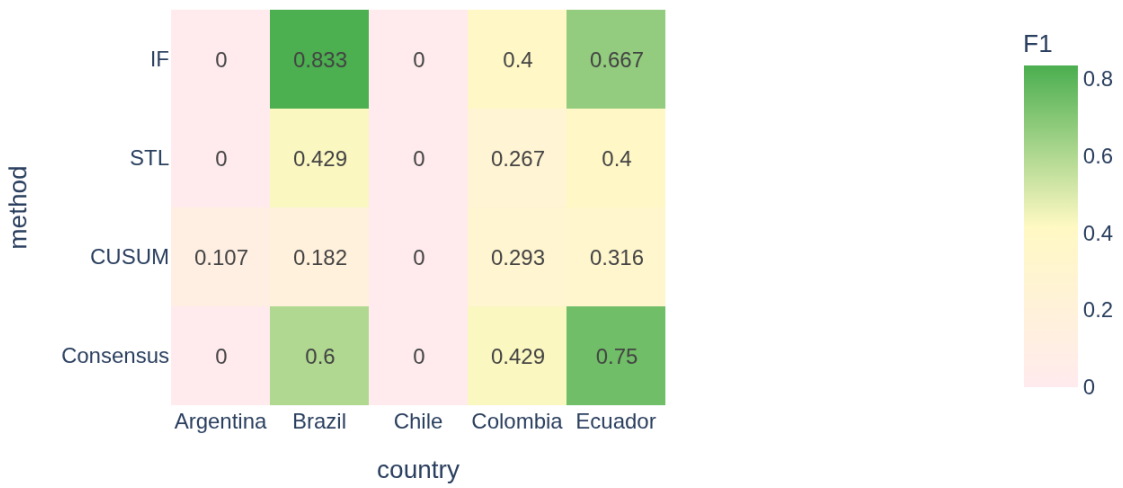
Consensus works in countries with hydro >30%. In fossil countries (Chile 14%, Argentina 14%), crises have different signatures (prices, heat) not detectable by STL/CUSUM.

Hydro Dependency vs F1-Score del Consenso (hallazgo clave)



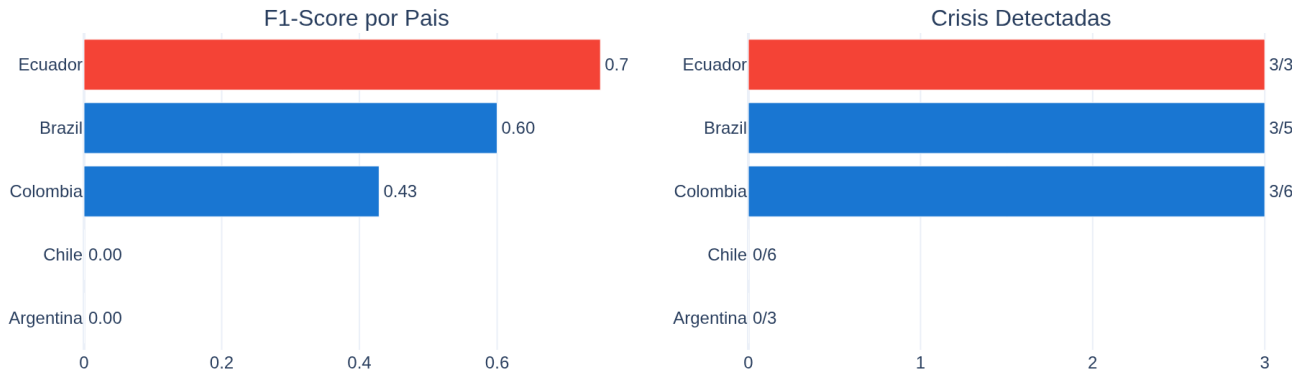
Hydro dependency vs F1

Cross-Country Validation: F1-Score by Method and Country

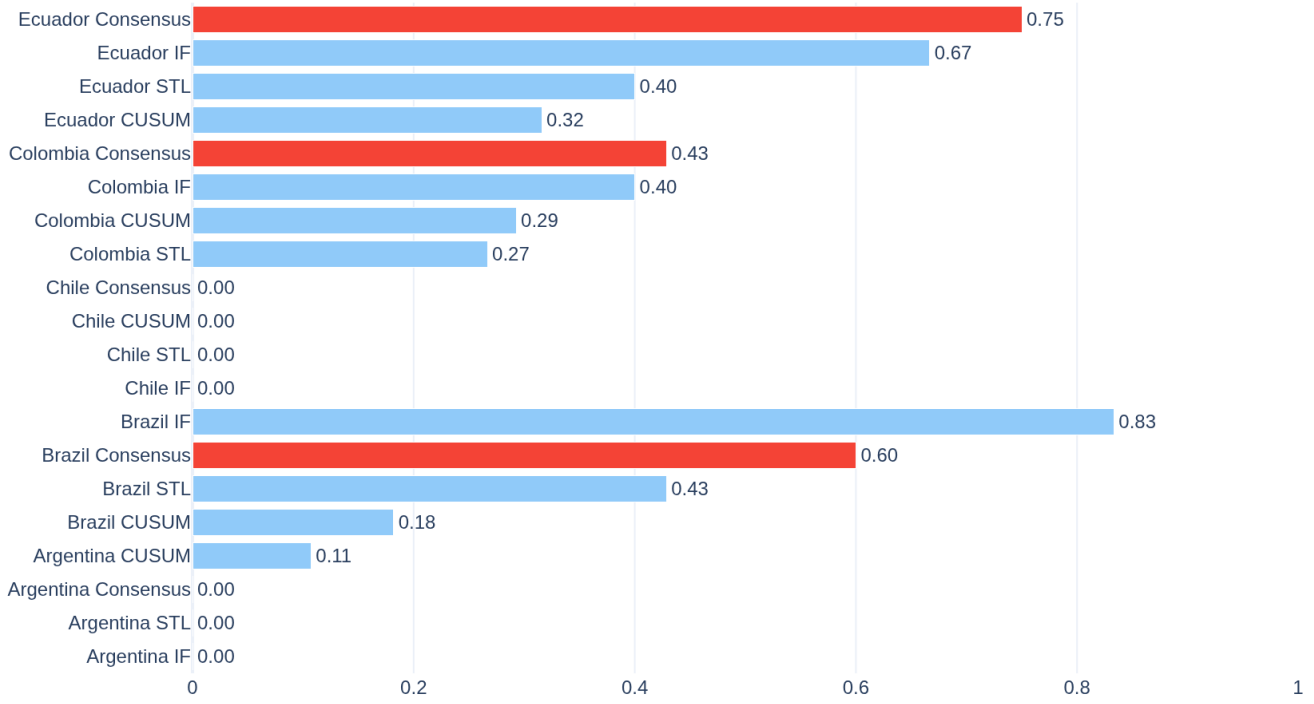


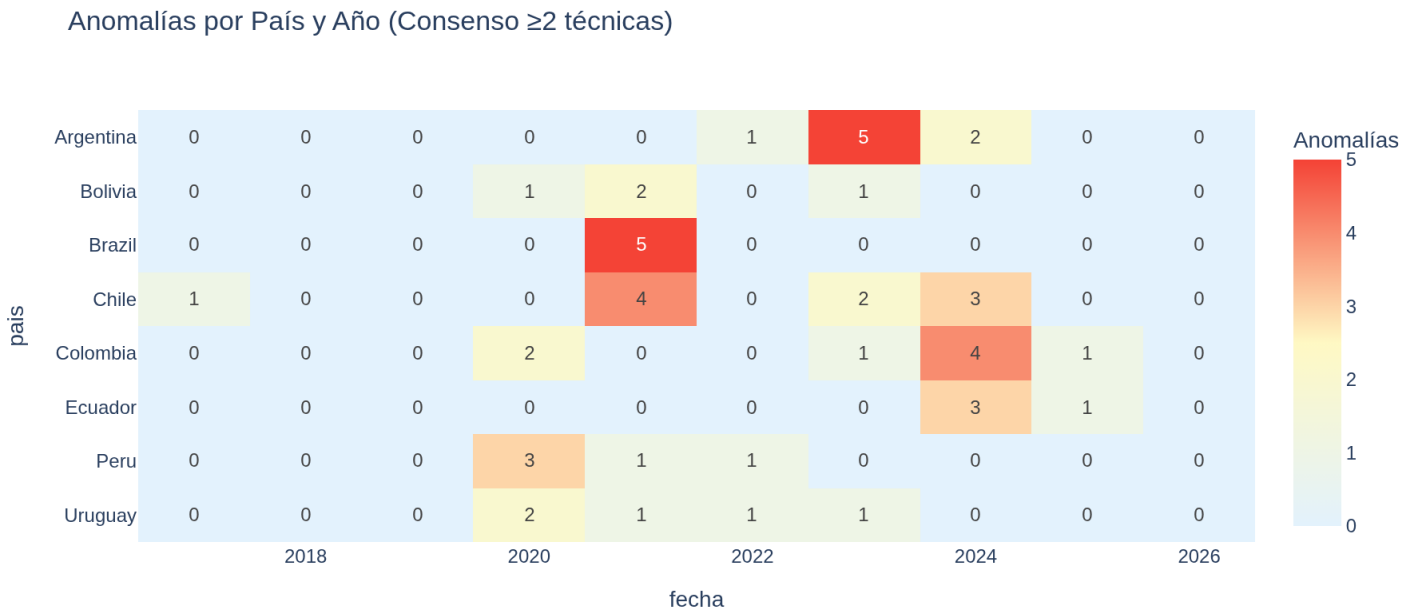
8. Cross-Country

Consenso Multi-Tecnica: Validacion Cross-Country



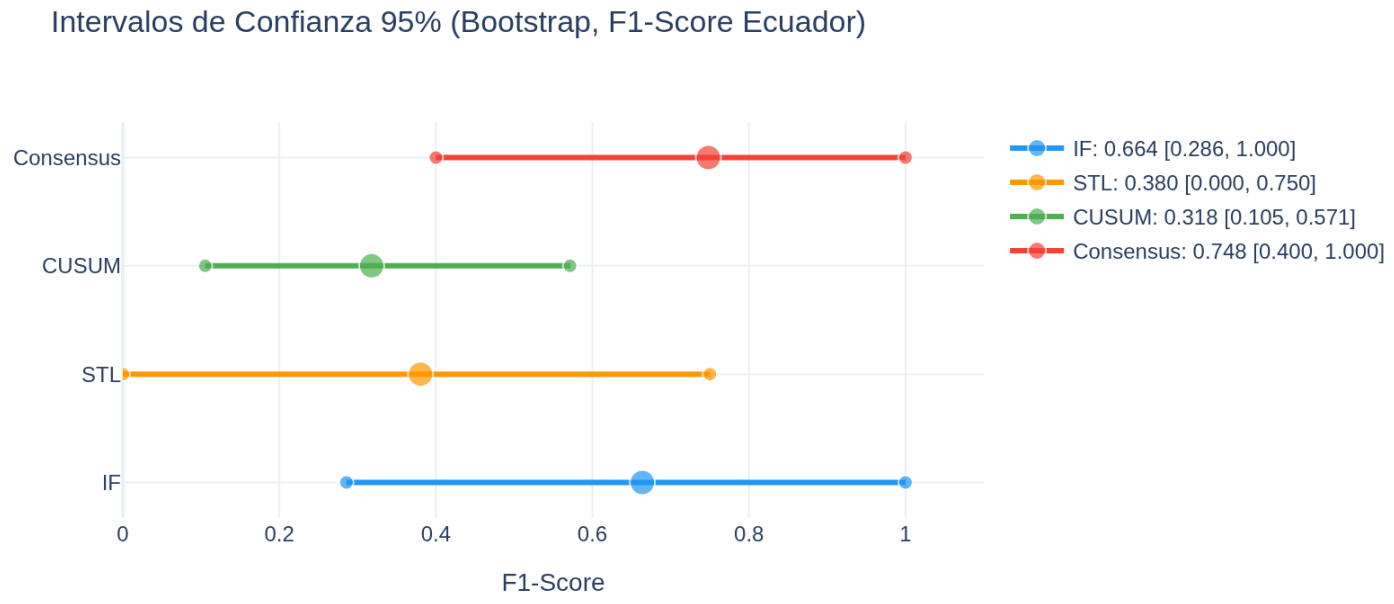
Consensus (red) vs Individual Methods by Country | Consensus wins 3/5



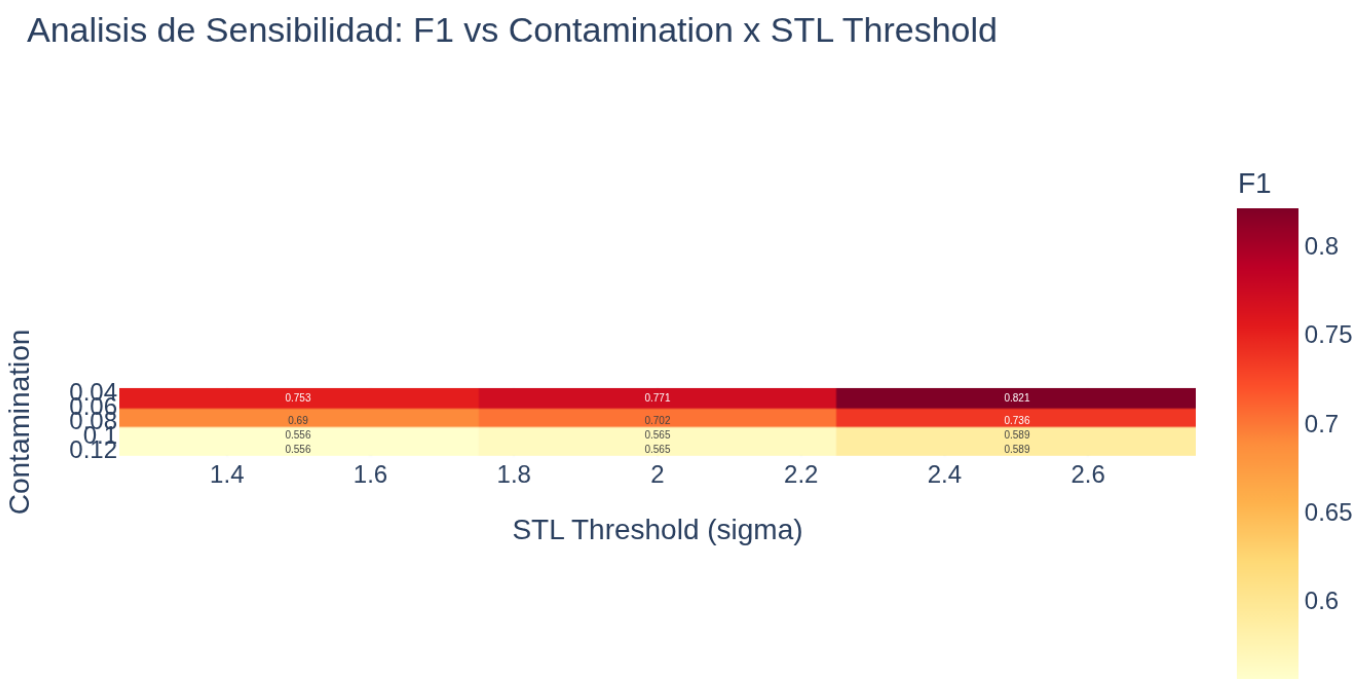

Anomalías por País y Año (Consenso ≥2 técnicas)
pais
Argentina 0 0 0 0 0 1 5 2 0 0
Bolivia 0 0 0 1 2 0 1 0 0 0
Brazil 0 0 0 0 5 0 0 0 0 0
Chile 1 0 0 0 4 0 2 3 0 0
Colombia 0 0 0 2 0 0 1 4 1 0
Ecuador 0 0 0 0 0 0 0 3 1 0
Peru 0 0 0 3 1 1 0 0 0 0
Uruguay 0 0 0 2 1 1 1 0 0 0
2018
2020
2022
2024
2026
fecha
Anomalías
5
4
3
2
1
0

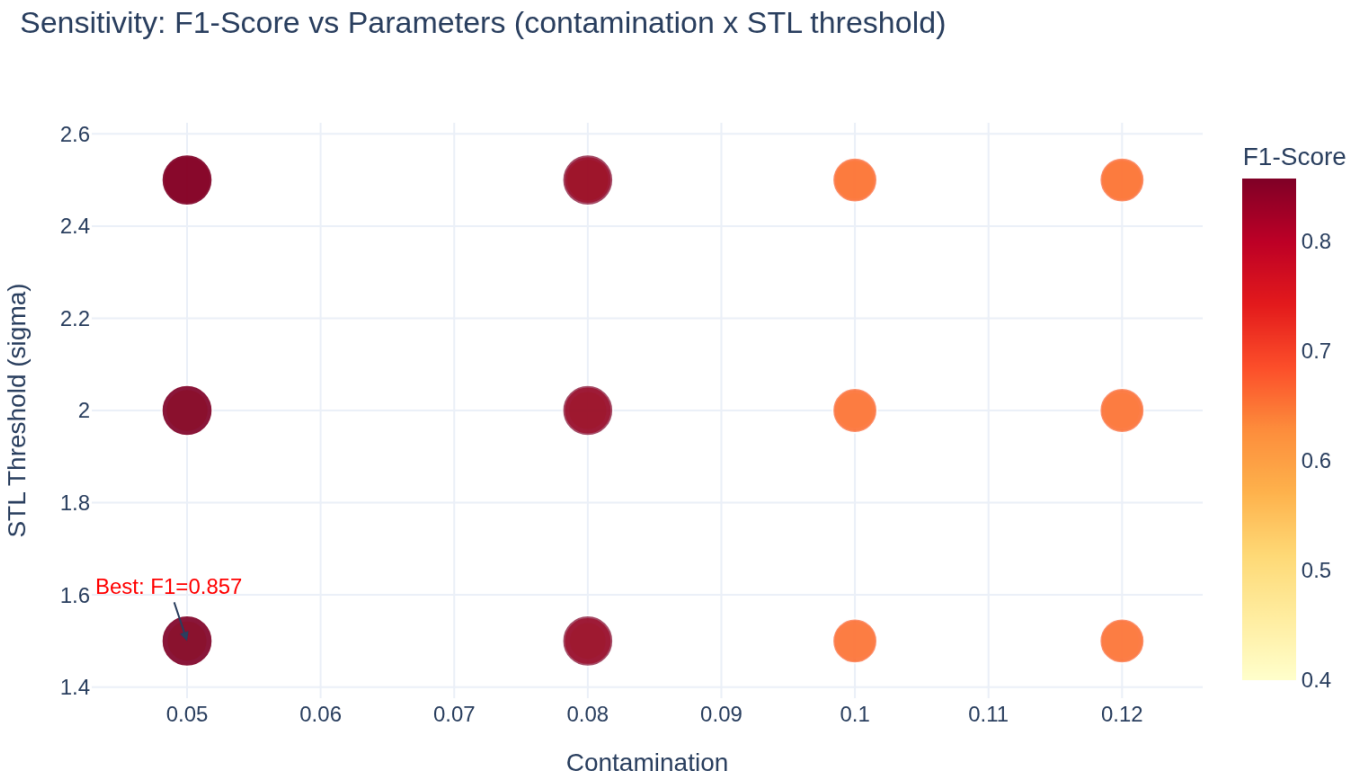
9. CIs + McNemar

Bootstrap 1000 iterations. McNemar consensus vs IF: p=1.0 (not significant at n=73). F1 difference 0.750 vs 0.667 exists but N lacks statistical power.

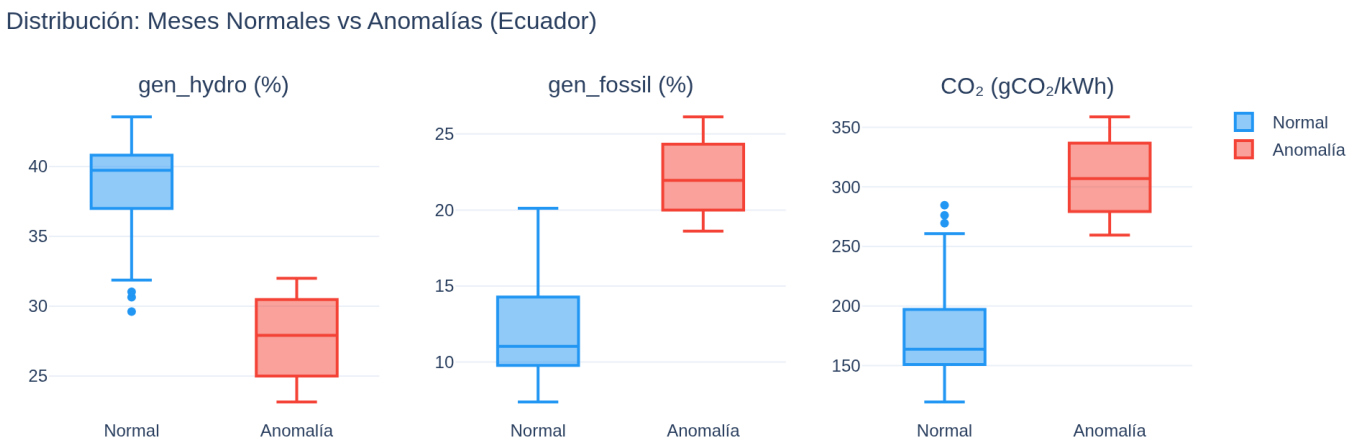


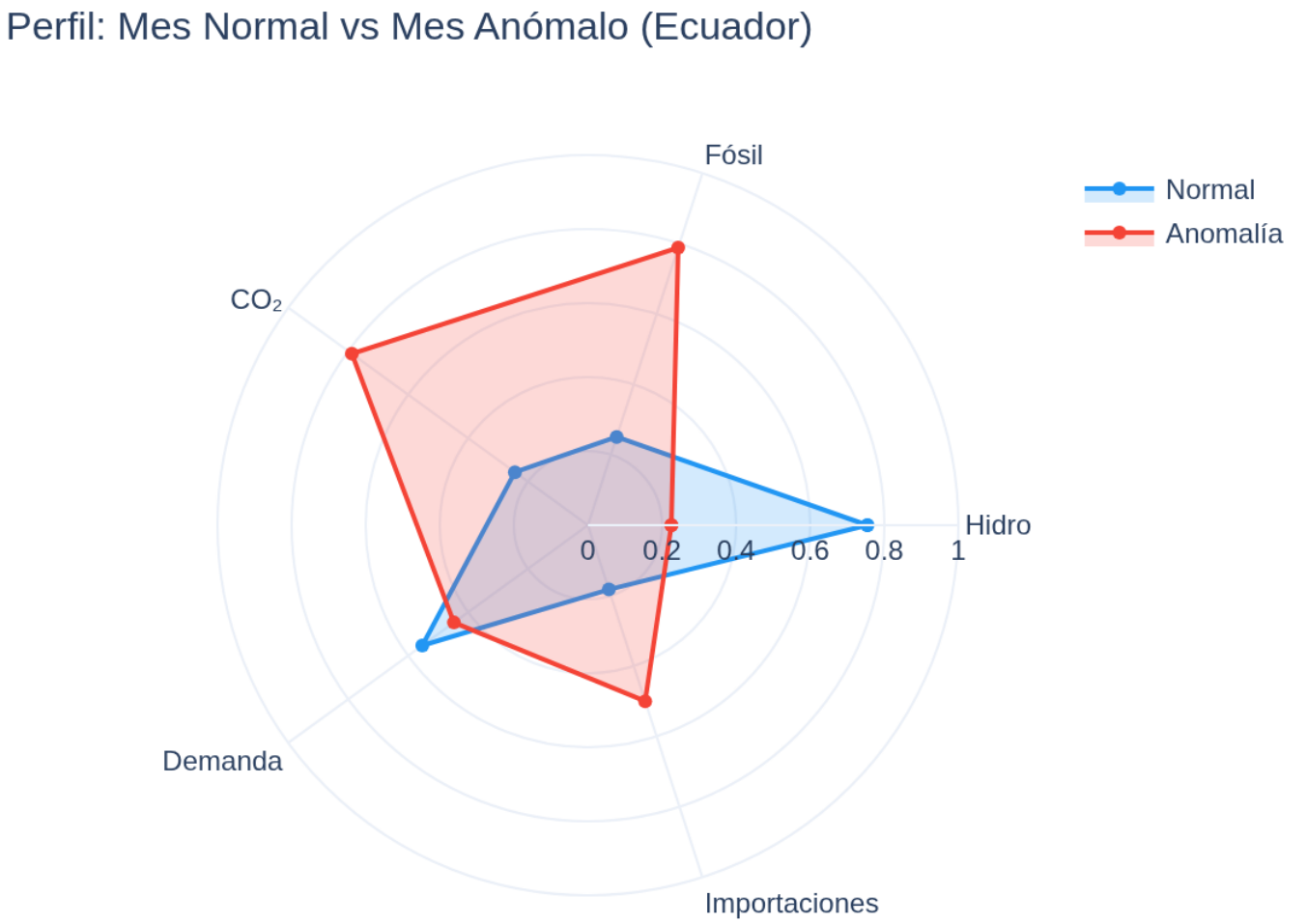
10. Sensitivity



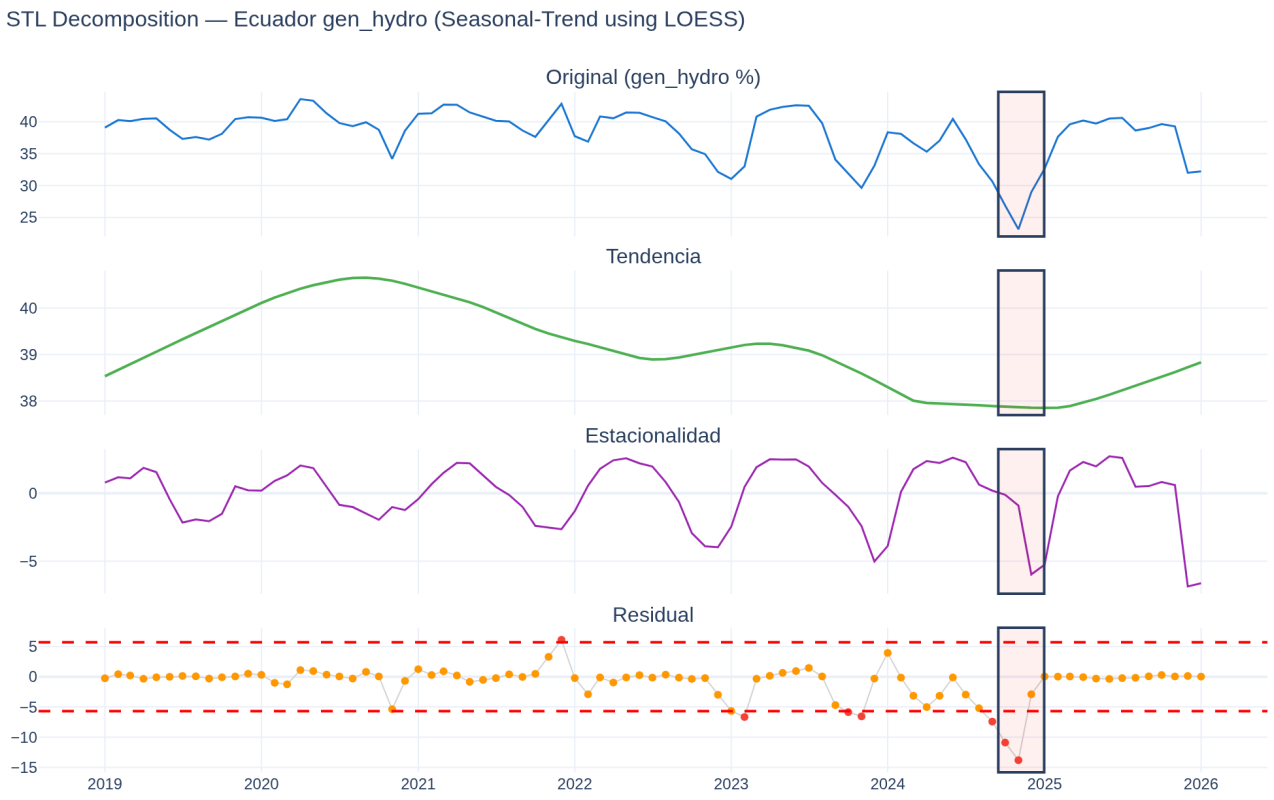

Sensitivity: F1-Score vs Parameters (contamination x STL threshold)
STL Threshold (sigma)
2.6
2.4
2.2
2
1.8
1.6
1.4
Best: F1=0.857
0.05
0.06
0.07
0.08
0.09
0.1
0.11
0.12
Contamination
F1-Score
0.8
0.7
0.6
0.5
0.4

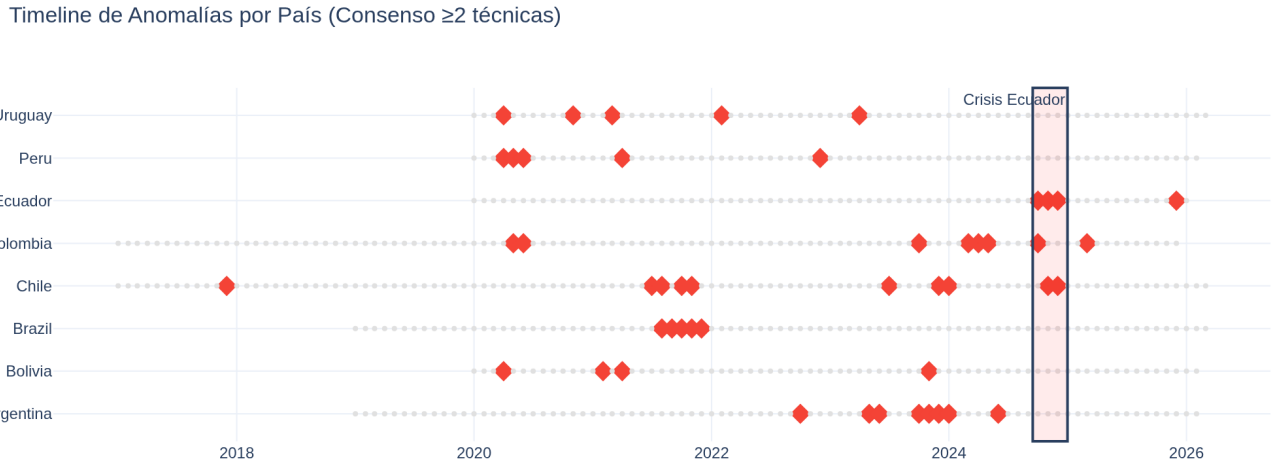
11. Validation





12. STL + Multi-Country





13. Limitations

1. McNemar p=1.0: cannot prove consensus > IF at n=73
2. Wide bootstrap CI: F1=[0.40, 1.00] due to small N
3. Fails in low-hydro countries (Chile F1=0, Argentina F1=0)
4. Broad GT (decree 9m) recall=33%: only detects 3 peak months
5. 85 months Ecuador is minimum viable
6. Monthly data smooths gradual crises

14. Applications

Grid operators (CENACE, XM, COES), regulators, traders, climate (El Nino), academic (multi-technique), policy (diversification).

Journals: Energy Policy, Applied Energy, IEEE Access, Energies (MDPI).

Suggested title: "Hydro-Weighted Consensus Anomaly Detection for Energy Crises in Latin American Power Grids"

15. References

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[3] Page (1954). CUSUM. Biometrika.
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